What Makes a Photo Popular?

An exploratory analysis of 25,000 photos from Unsplash Lite

Omar Saad Saleh Alshuail

Table of Contents

1	Title Page	1
2	Table of Contents	2
3	Introduction	3
4	Data Exploration	4
5	Data Cleaning	5
6	Exploratory Data Analysis	6
7	Dashboard Overview	10
8	Insights & Conclusion	12

3 Introduction

Dataset	Unsplash Lite Dataset
Source	github.com/unsplash/datasets
Size	25,000 photos, 31 columns

Description: free photos published by photographers on Unsplash. Each photo has the camera settings it was taken with (EXIF), its size, its upload date, and how many times it was viewed and downloaded.

Why this dataset: I work in photography, and this data links camera settings to how people react to a photo, so I can tell the story from a photographer's point of view.

Questions

- Do phone photos get downloaded as much as camera photos?
- Which camera brands are used most?
- Do landscape or portrait photos get downloaded more?
- Do a few photos take most of the downloads?
- Can a simple model predict which photos get downloaded more?

Expected result

I expected cameras to clearly beat phones. The data showed the opposite.

Metric

Download rate = downloads ÷ views. I used it instead of raw downloads because older photos collected more downloads just by being online longer.

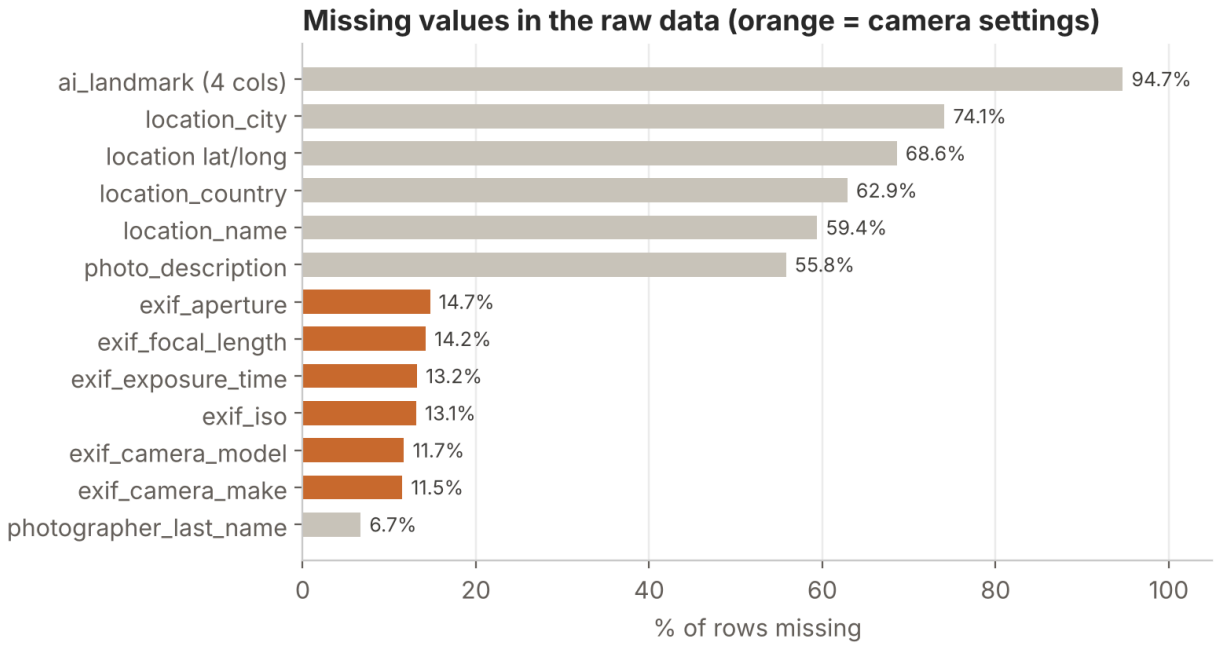
4 Data Exploration

- Shape: `(25000, 31)`
- Types: 20 text, 7 float, 4 integer. Camera settings are stored as text in mixed formats, such as `"5,6"`, `"f/5"`, `"50mm"` and `"1/250"`.
- No duplicate rows.

Summary statistics

	views	downloads	exif_iso
Mean	4,426,662	20,979	562
Median	841,095	4,646	200
Min	13,867	129	0
Max	2,732,293,685	4,929,347	51,200

The mean is far above the median, so the data is heavily skewed. An ISO of 0 is impossible.



Missing values: location data is missing for most photos, camera settings for 11% to 15% of them.

5 Data Cleaning

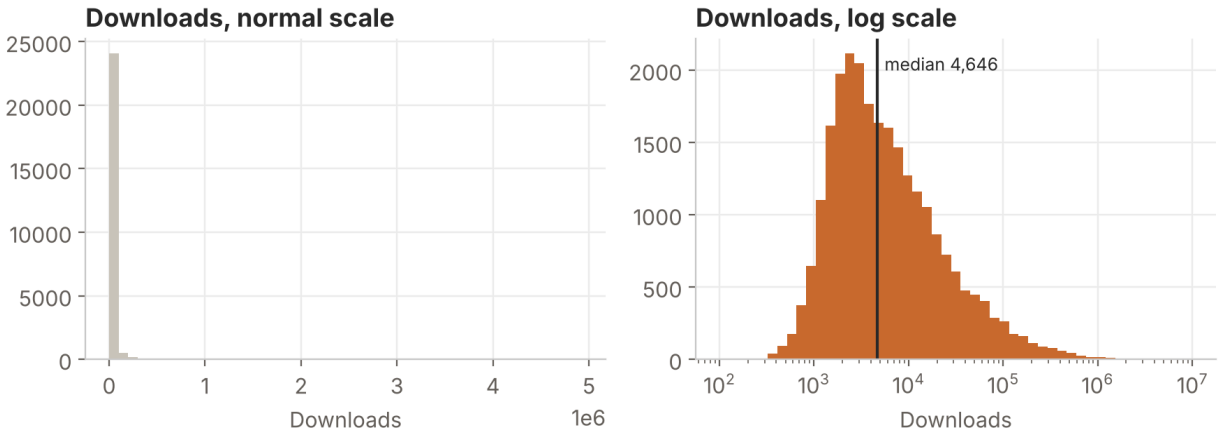
1. Kept 19 useful columns out of 31.
2. Renamed columns to shorter names, such as `aperture` and `downloads`.
3. Converted camera settings to numbers: `"5,6"` to 5.6, `"50mm"` to 50, and `"1/250"` to 0.004 seconds.
4. Turned impossible values into missing values: ISO 0, apertures outside f/0.95 to f/32, and a 0 mm focal length.
5. Filled missing brands with `"unknown"`. I did not fill ISO or aperture with the mean, so no fake settings were invented.
6. Merged brand spellings: `"NIKON CORPORATION"` and `"Nikon"` became `nikon`.
7. Added new columns: device type, orientation, year, download rate and megapixels.
8. Saved the result as `data/clean_data.csv`.

Before and after

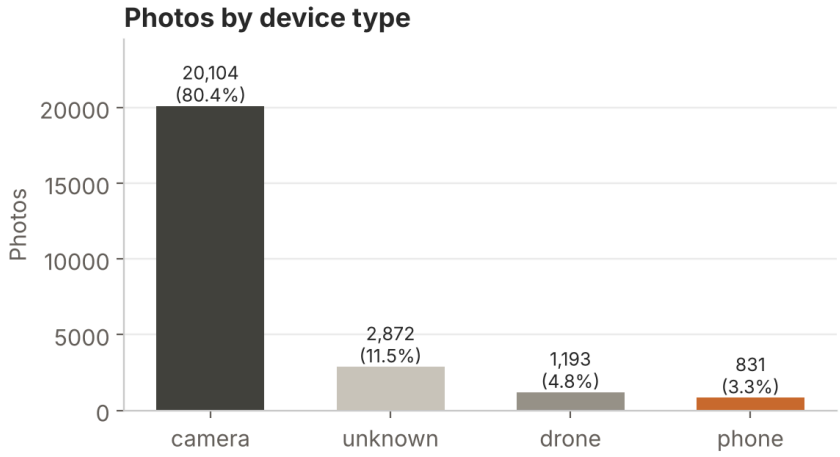
	Before	After
Rows	25,000	25,000
Columns	31	26
Camera settings	Mixed text	Numbers

6 Exploratory Data Analysis

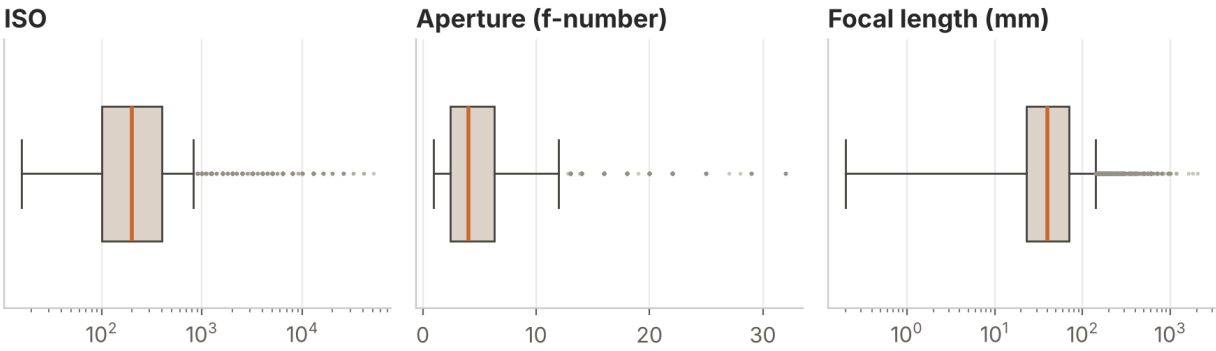
Univariate



Downloads are heavily skewed: the typical photo gets 4,646 downloads, and the top 1% of photos take 30% of all downloads.

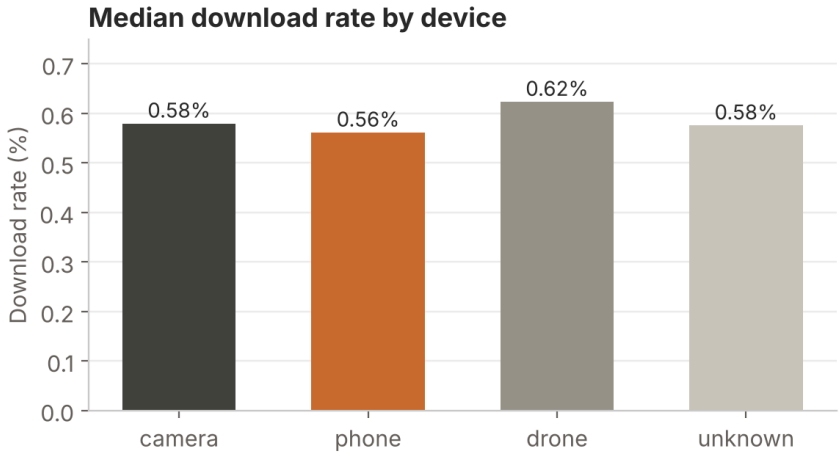


80% of photos were taken with cameras and only 3% with phones.

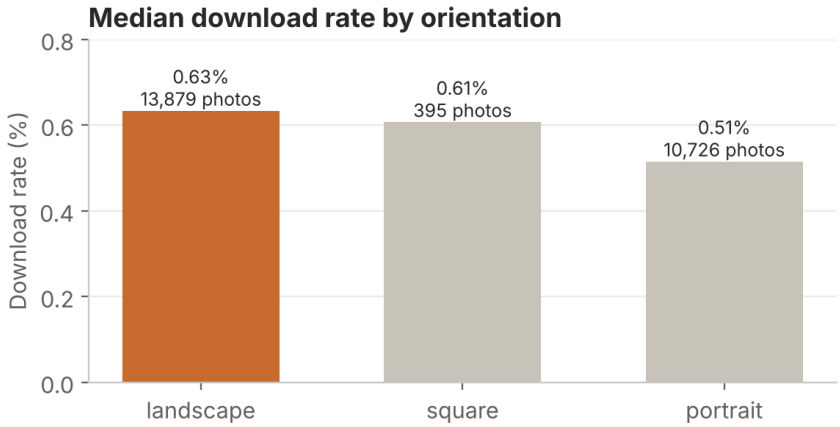


The most common settings are ISO 100, f/2.8 and 50 mm. The far points are real shots (night photography and telescopes).

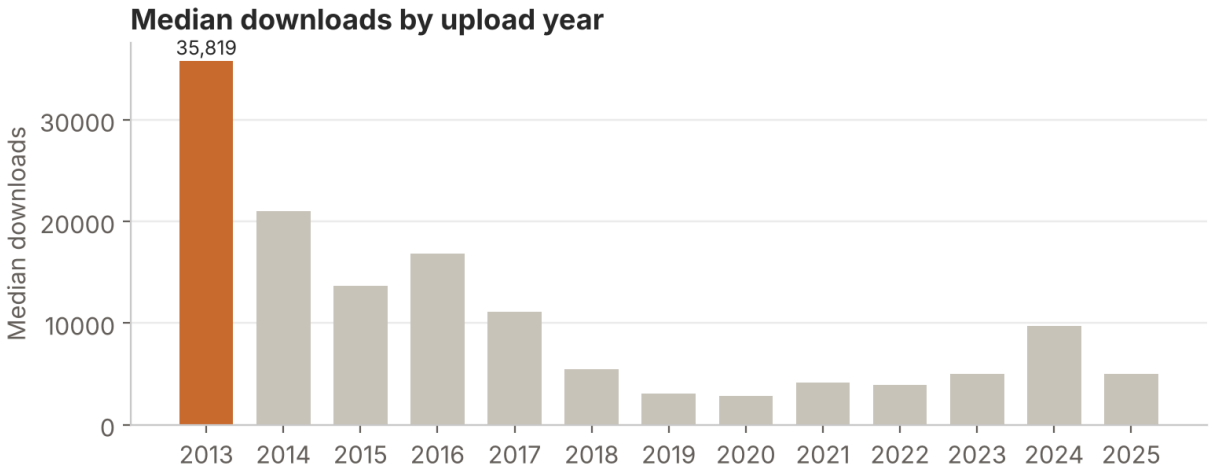
Bivariate & multivariate



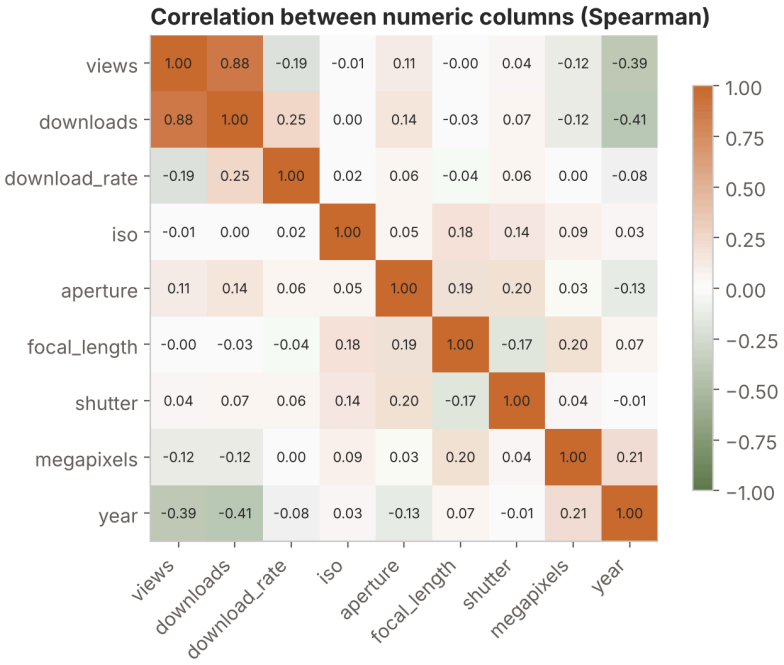
Download rate is almost the same for every device: phones 0.56%, cameras 0.58%.



Landscape photos have a download rate about 23% higher than portrait photos.



Older photos collected more downloads because of time online, which is why download rate is used.

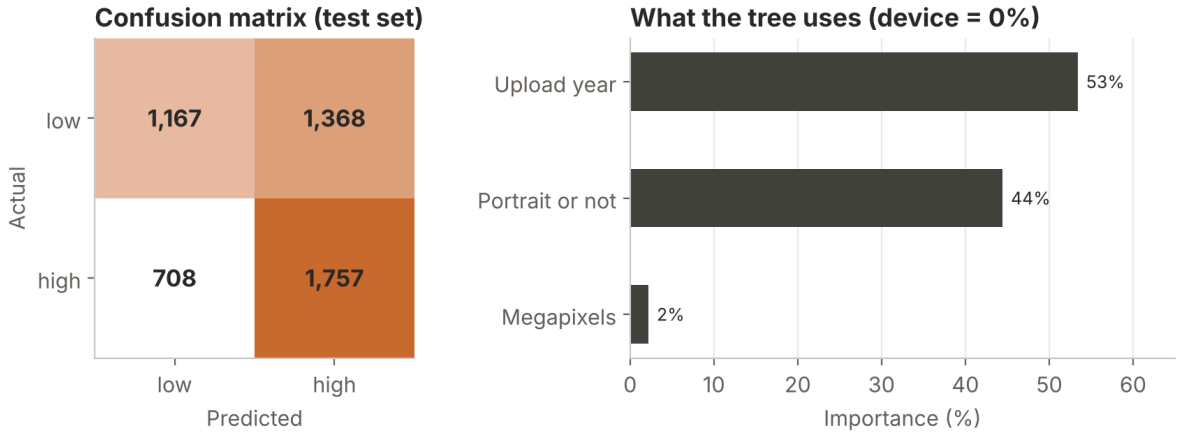


Camera settings barely relate to download rate. Views and downloads are strongly related (0.88).

Predictive model

- **Model:** Decision Tree (`max_depth=4`).
- **Target:** is the photo's download rate above the median?
- **Inputs:** orientation, device type, upload year and megapixels.

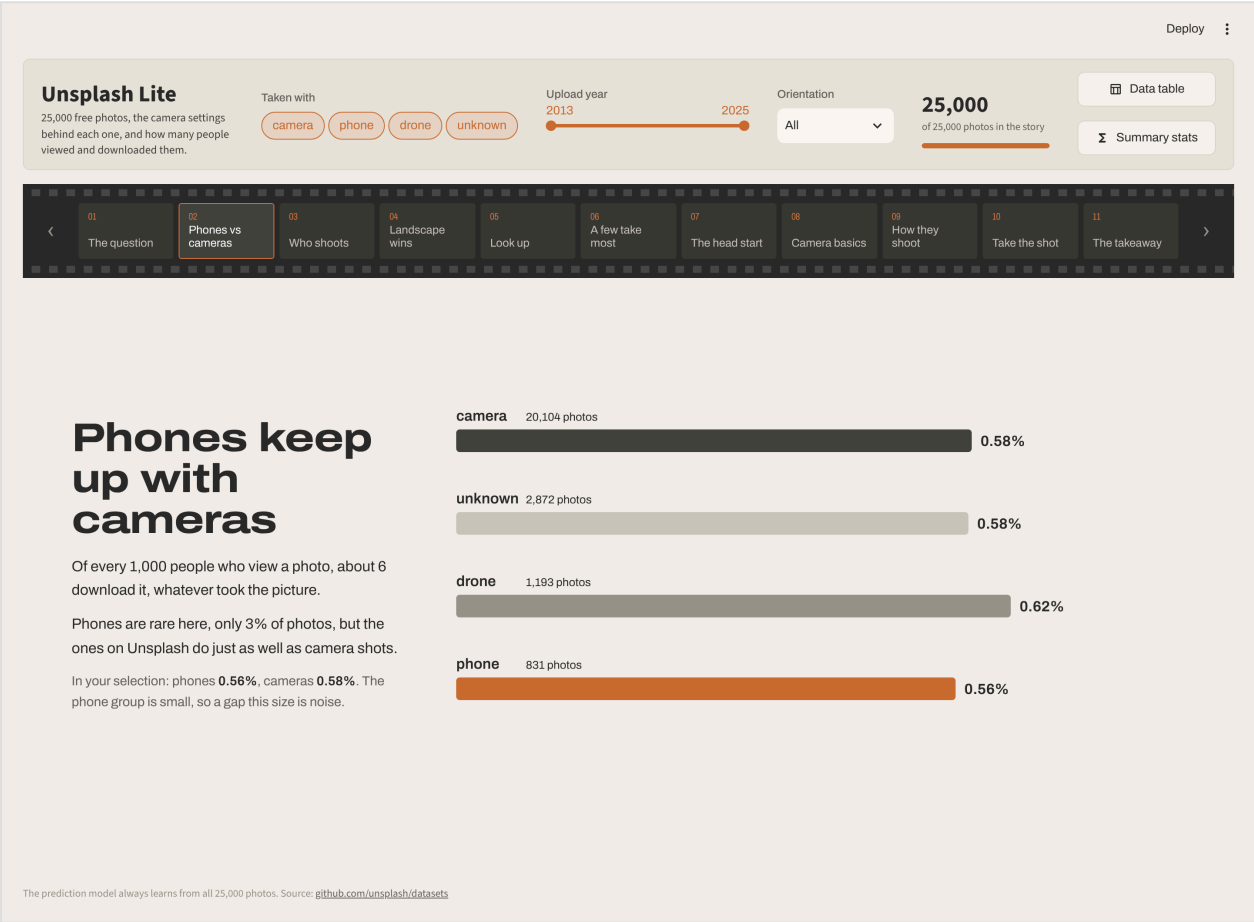
Metric	Result
Test accuracy	58.5%
Train accuracy	59.1%
5-fold cross-validation	58.7%
Random guess	50%



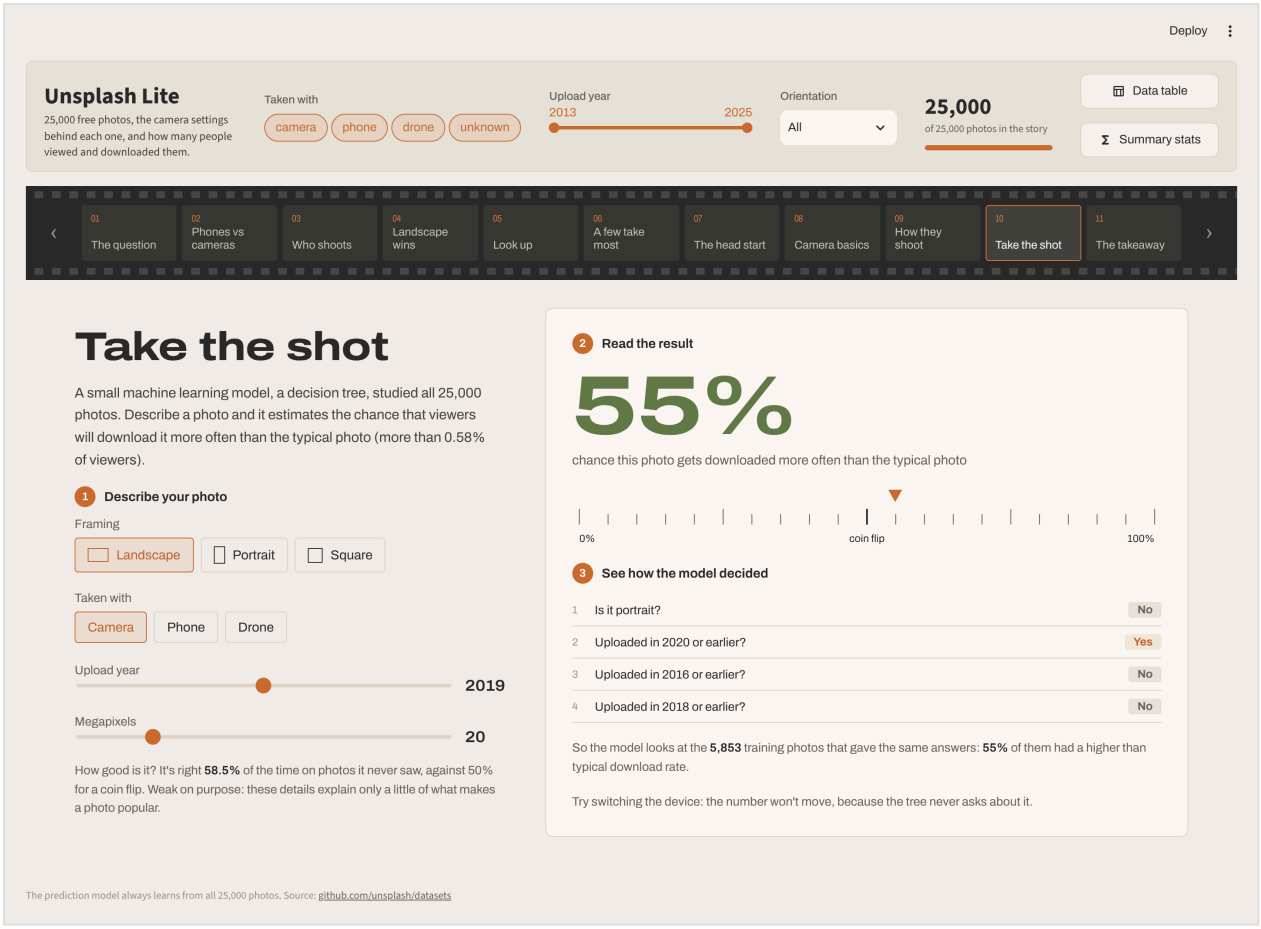
The model only uses upload year and orientation. Device type has zero importance.

7 Dashboard Overview

- Built with Streamlit and deployed on Streamlit Community Cloud.
- Top bar: dataset description and filters (multiselect, slider, dropdown).
- Main page: 11 frames explaining the findings, including an interactive model, plus buttons that open a data preview and summary statistics.
- Code: github.com/oShuail/EDA-unsplash-dataset



The "Phones vs cameras" frame with the filters in the top bar.



The model frame: the user describes a photo, and the model shows its prediction and how it decided.

8 Insights & Conclusion

Main insights

- The device does not decide success: phones 0.56%, cameras 0.58%.
- Landscape photos get downloaded more than portrait photos.
- Popularity is unequal: the top 1% of photos take 30% of all downloads.
- Canon, Nikon and Sony took 78% of the photos that have camera info.

Outliers

- One photo has 2.7 billion views, another 4.9 million downloads.
- Extreme but real settings (ISO 51,200, a 3 hour shutter) were kept.

Conclusion

You don't need an expensive camera for a photo to succeed. Orientation and reaching viewers matter more.

What I learned

- Missing values can be information in themselves.
- Choosing the right metric can change the conclusion.
- Skewed data needs the median and log scales.

Limitations

- One curated platform, mostly nature photos.
- Few phone photos (831).
- 43% of photos are from 2019 alone.

Recommendations

- Add the colors and keywords tables to study photo content.
- Measure downloads per day since upload.
- Use the full version of the dataset.